

Zero Reflection Symmetry: The First Analytic Material of the Riemann Bridge (Exercise XXVI)

Anonymous Author(s)

Abstract

We report the first analytic theorem of the Riemann-direction chain: the zero reflection symmetry of the Riemann zeta function, proved from the classical functional equation in Lean 4 / mathlib with zero sorry. The verification map of the Riemann hypothesis has two completed banks — the Euler-product bank on $\text{Re}(s) \geq 1$ (zero-free region) and the geometric bank (critical line $\leftrightarrow$ critical circle) — and one open span: the analytic control of zeros in the critical strip $0 < \text{Re}(s) < 1$. The functional equation $\zeta(1-s) = 2(2\pi)^{-s}\Gamma(s) \cos(\pi s/2)\zeta(s)$, present in mathlib as `riemannZeta_one_sub`, is the only known analytic material that connects the two banks. This exercise formalizes its first corollary: every non-trivial zero reflects to a zero ($\zeta(s) = 0 \Rightarrow \zeta(1-s) = 0$), the reflection stays in the strip, and an off-line zero ($\text{Re}(s) \neq 1/2$) therefore occurs in pairs. Three theorems, zero errors, mathlib v4.32.2.

Keywords: Lean, formalization, Riemann zeta function, functional equation, zero symmetry, 0pat

1 The position of this exercise

The Riemann-direction chain is a verification map with two completed banks and one open span. The left bank is analytic: the Euler product converges on $\text{Re}(s) > 1$ and ζ has no zeros on $\text{Re}(s) \geq 1$ (Exercise C025). The right bank is geometric: the critical line is the critical circle under inversion, and the hypothetical zero positions and the critical circle are the same object (Exercises C019–C022). The open span is the critical strip $0 < \text{Re}(s) < 1$: no theorem of the chain controls the positions of the zeros of ζ there.

The only known analytic material that spans the two banks is the functional equation. Mathlib states it in classical form (`riemannZeta_one_sub`): for $s \neq -n$ ($n \in \mathbb{N}$) and $s \neq 1$,

$$\zeta(1-s) = 2(2\pi)^{-s} \Gamma(s) \cos(\pi s/2) \zeta(s).$$

This exercise takes the first corollary of this equation: zero reflection symmetry. It is the first theorem of the chain that genuinely uses analysis — the functional equation — rather than algebra or geometry.

2 The theorems

Three theorems, all machine-checked with zero sorry.

Reflection (`zeta_zero_reflection`). If $\zeta(s) = 0$ and s is not a trivial zero position ($s \neq -n$ for all n) and not the pole ($s \neq 1$), then $\zeta(1-s) = 0$. The proof is one rewriting step: substitute the functional equation and multiply the zero by

the coefficient. The coefficient need not be nonzero — zero times any coefficient is zero — so no analytic nonvanishing is required.

The reflection stays in the strip (`zeta_zero_strip_reflection`).

If $0 < \text{Re}(s) < 1$, then $0 < \text{Re}(1-s) < 1$ as well. Together with reflection: a zero in the strip reflects to a zero in the strip. Zeros of ζ are therefore symmetric about the critical line $\text{Re}(s) = 1/2$ within the strip.

Off-line zeros come in pairs (`zeta_zero_offline_pair`).

If $\text{Re}(s) \neq 1/2$ (the zero is off the critical line), then $1-s \neq s$, and the reflection is a *different* zero of the strip. The proof uses the real-part computation: $1-s = s$ would force $\text{Re}(s) = 1/2$. This is the first structural constraint on off-line zeros that the chain obtains: if an off-line zero exists, it is accompanied by a second, distinct off-line zero. The Riemann hypothesis asserts that no such pair exists; this theorem asserts only that pairs are the only possible form of violation.

3 Honest boundary and position in the map

As in every exercise of the series: the content is classical mathematics (“novelty_status: KNOWN” — the reflection symmetry is a standard corollary of the functional equation, known since Riemann), the proof is machine-checked, and no claim is made of proving the Riemann hypothesis. The value of the exercise is positional: it is the first analytic node of the map’s open span, it connects the two completed banks through the functional equation, and it converts the question “is there an off-line zero?” into the more structured question “is there an off-line zero pair?”. The bridge now has its first material; the span remains open.

4 Formalization

The proof is a single Lean file (`proof.lean`, 62 lines), importing only `Mathlib.NumberTheory.LSeries.RiemannZeta.Complex` compiled with Lean 4.32.2 / mathlib v4.32.2, zero errors, zero sorry. The functional equation `riemannZeta_one_sub` is mathlib’s classical statement; `Complex.sub_re` and `linarith` handle the strip inequalities.

Theorem	Statement
<code>zeta_zero_reflection</code>	$\zeta(s) = 0 \Rightarrow \zeta(1-s) = 0$
<code>zeta_zero_strip_reflection</code>	strip-preserving reflection
<code>zeta_zero_offline_pair</code>	$\text{Re}(s) \neq 1/2 \Rightarrow 1-s \neq s$ and zero

5 Conclusion

The first analytic material of the Riemann bridge is in place: zero reflection symmetry, proved from the functional equation, machine-checked in Lean. The map now reads: left bank (Euler product, zero-free region), first span material (reflection symmetry, off-line pairs), right bank (critical line $\leftrightarrow$

critical circle). The span remains open — the reflection symmetry permits off-line pairs and the Riemann hypothesis denies them — but the question it poses is now exactly structured: can the analysis show that no off-line zero pair exists?

References